Sub Code:EC303MTU

Paper Id: 130306 Roll No:

B TECH
(SEM-III) THEORY EXAMINATION 2019-20
SIGNALS & SYSTEMS

Time: 3 Hours ***Total Marks: 100***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

a.	Plot the graph of $x(t) = u(t + 1/2) + \delta(t+3)$
b.	Check whether the given system are Time Variant and Causal **y(t)= x(-t).**
c.	Write down the Final value condition related to Z- transform.
d.	Distinguish between Random & Deterministic signal.
e.	Calculate the Laplace Transform of the signal of **x(t)=u(-t).**
f.	Calculate the Energy of the signal $x(t) = e^{-t}u(t)$
g.	Write the condition to check the Stability of any LTI System.
h.	Define signal.
i.	Determine the DTFT of the signal $x(n)=\delta(n)$
j.	Prove that system **$y(t)= x^2(-t)$** is non-linear.

SECTION B

2. Attempt any *three* of the following: 10x3=30

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a.	Check whether following systems are causal, linear and time invariant: **i) $y(n) = x^2(n+4)$** **ii) y(t)= x(t-3)**
b.	Derive the time response of the second order system subjected to unit step input.
c.	For the Laplace transform of $x(t) = e^t sin(2t)\ t \leq 0$.Indicate the location of its poles and its region of convergence.
d.	Find the inverse *z* transform of $X(z)= \frac{(z-\frac{1}{2})}{(z-1)(z-\frac{1}{3})}\ |z| > \frac{1}{3}$
e.	Determine the Laplace transform & find out ROC for **$x(t)=e^{-t}u(t)+e^{-4t}u(t)$** & depict the ROC and the location of poles and zeros in the s-plane.

SECTION C

3. Attempt any *one* part of the following: 10x1=10

a.	A causal LTI system is described by difference equation. y(n) = y(n-1) +2 + x(n-1) Find the system function H(z) for this system. Plot the poles zeros of H(z)
b.	Determine and sketch even and odd components for $x(t) = e^{-t}u(t)$

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4. Attempt any *one* part of the following: 10x1=10

a.	Determine the fundamental period of the following signals. i) **x(t)= 2cos (10t+1) – sin(4t-1)** ii) **x(t)= 1+ $e^{j4\pi n/7}$**
b.	Calculate the Z-Transform of $h_1(n)$.sketch its ROC. $h_1(n) = \left(1/3\right)^n u(n)$

5. Attempt any *one* part of the following: 10x1=10

a.	Find the DTFT of the causal sequence **$x(n)= a^n u(n)$, $a < 1$.** Derive the Parseval's theorem related to Continuous Time Fourier Transform.
b.	Calculate the Fourier transform of $x(t) = e^{-\tau}u(t)$ Also Sketch its spectrum.

6. Attempt any *one* part of the following: 10x1=10

a.	Determine the Inverse Laplace Transform f(t) of $F(s) = \frac{1}{(s+2)(s+4)(s+3)}$
b.	Calculate the Laplace transform of $x(t) = e^{-at} \sin \omega_o t$

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7. Attempt any *one* part of the following: 10x1=10

a.	Find the convolution y(n)= x(n)*h(n) where $x(n)=(\frac{1}{2})^n u(n)$, $h(n)=u(n)$
b.	Explain system bandwidth and rise time for low pass filter and prove that t_r =0.35/B.

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